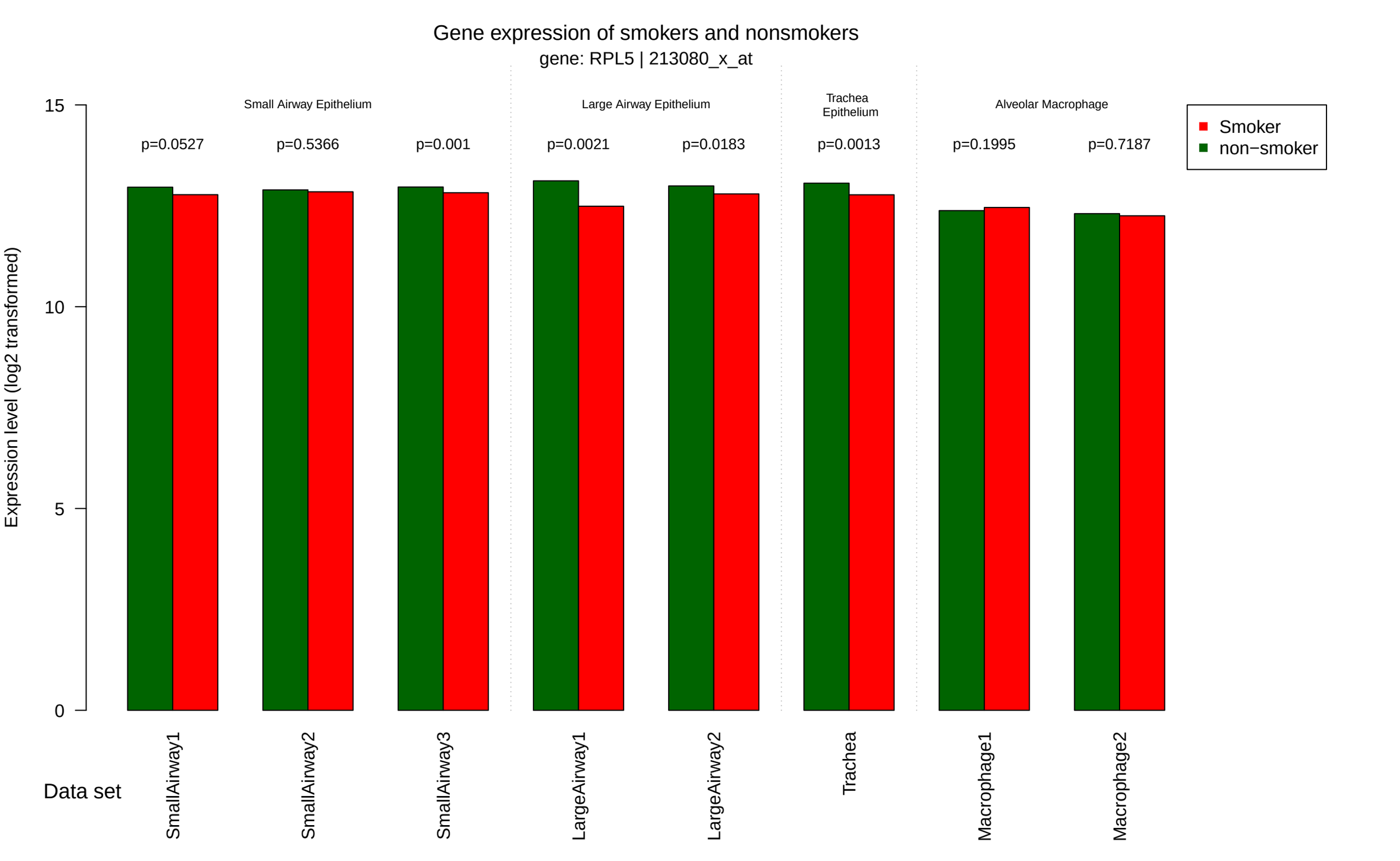
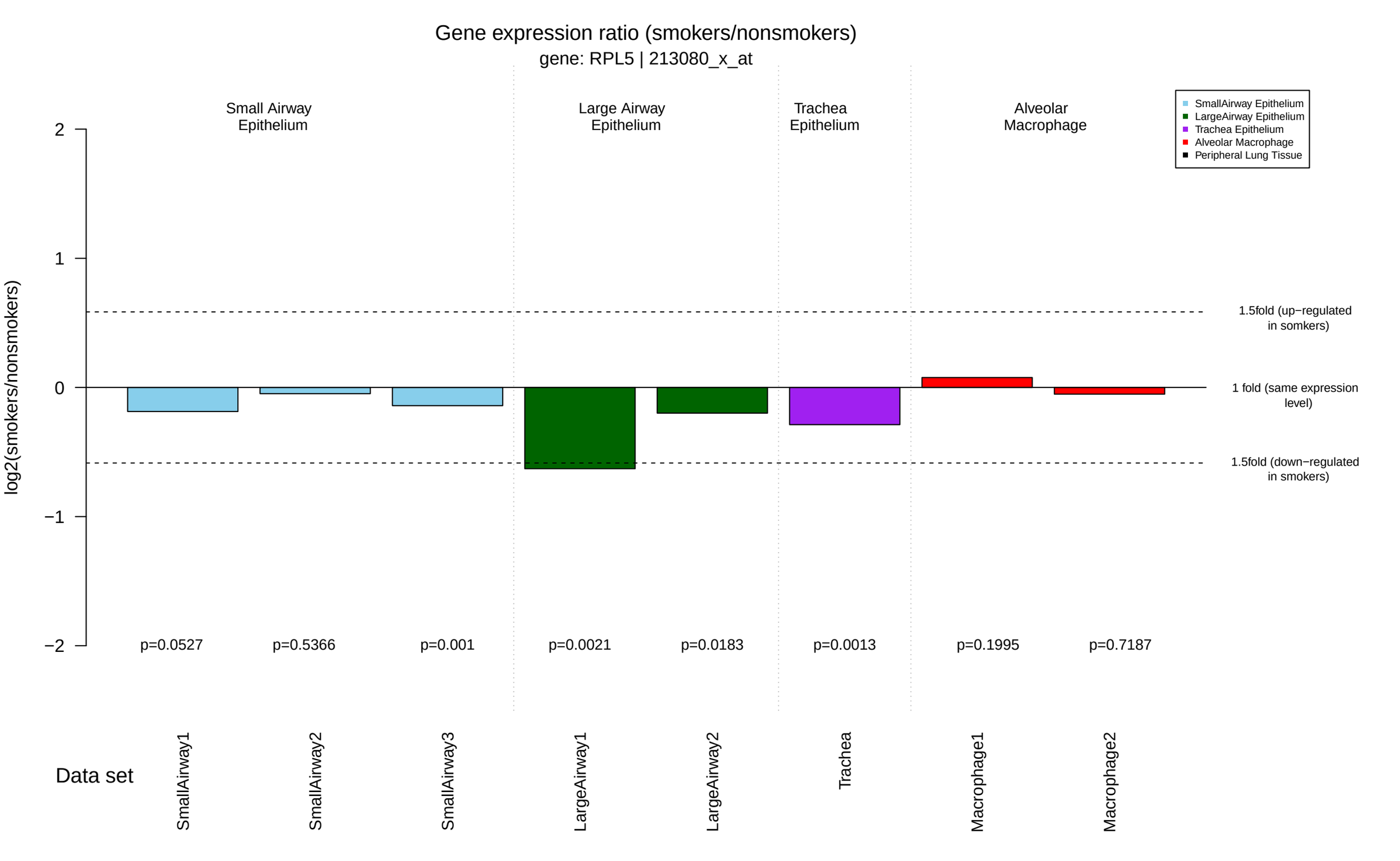
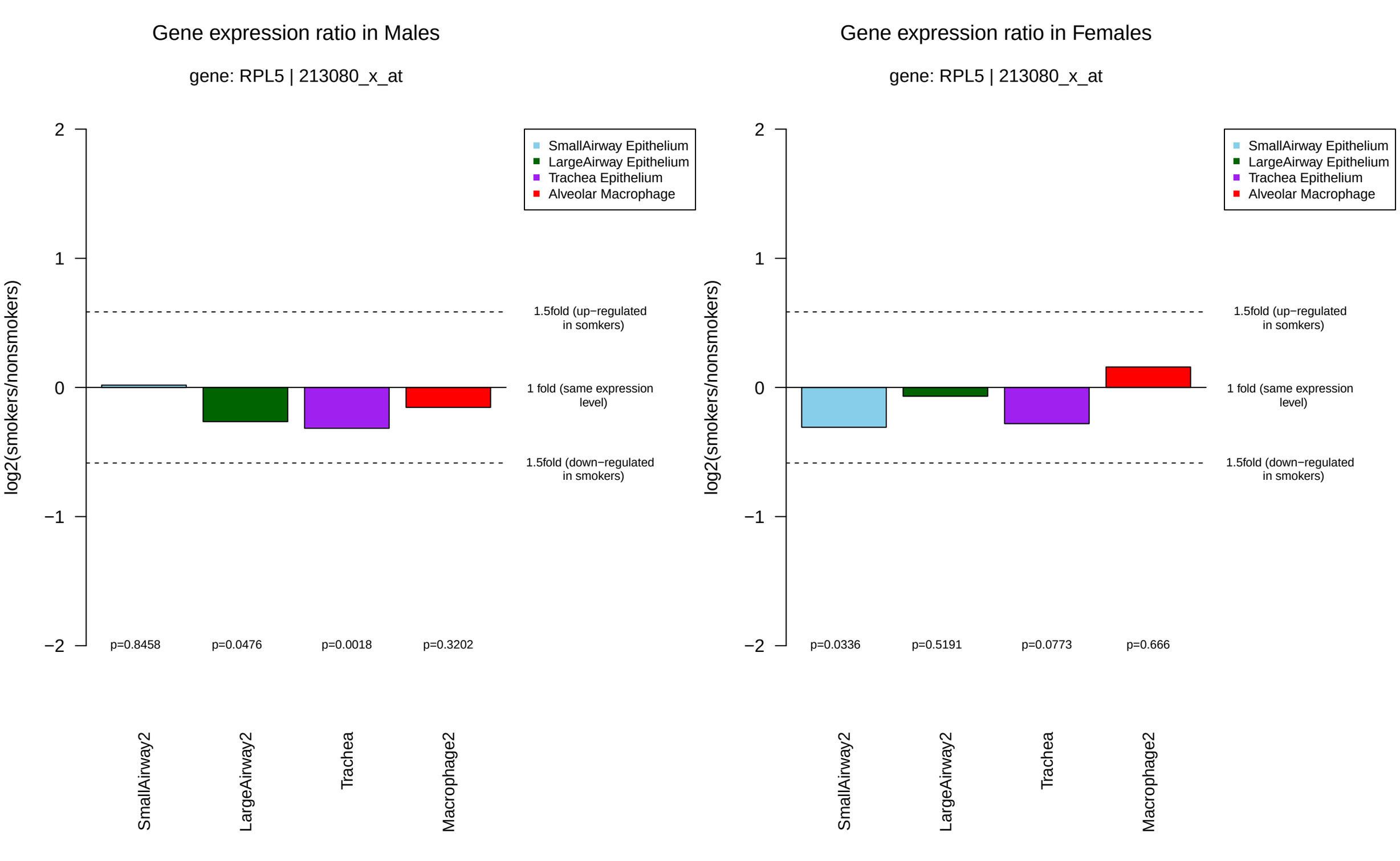
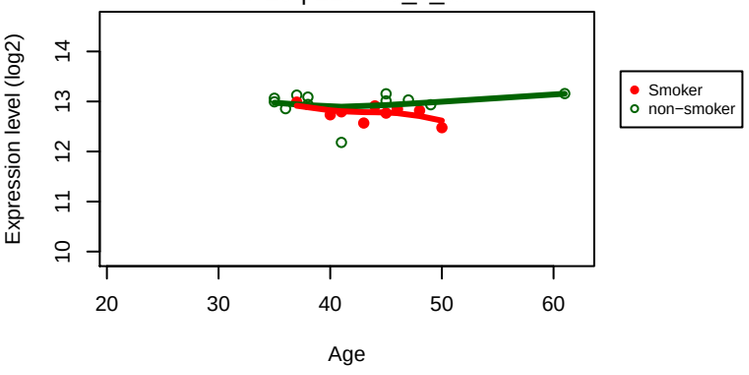
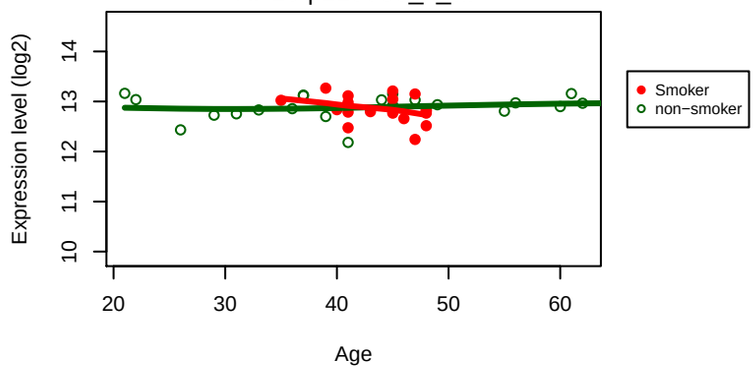
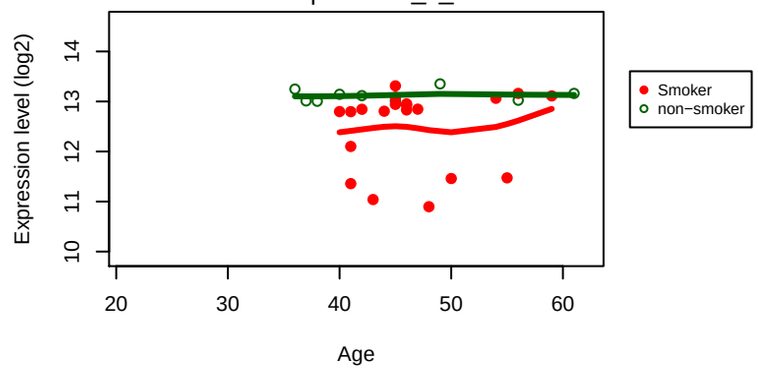

Gene expression of smokers and nonsmokers
gene: RPL5 | 213080_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0527
p=0.5366
p=0.001
p=0.0021
p=0.0183
p=0.0013
p=0.1995
p=0.7187
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

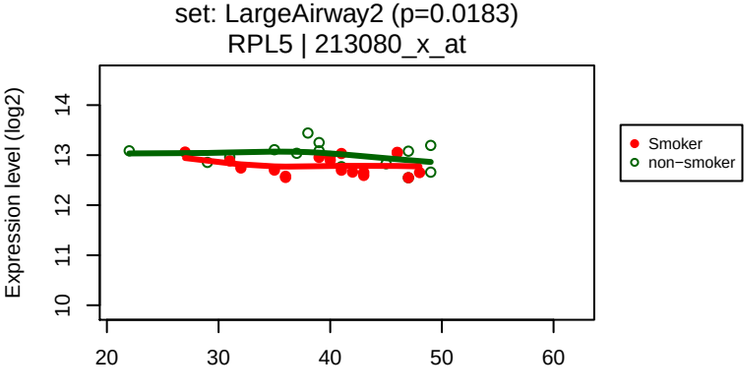

Gene expression ratio (smokers/nonsmokers)
gene: RPL5 | 213080_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0527
p=0.5366
p=0.001
p=0.0021
p=0.0183
p=0.0013
p=0.1995
p=0.7187
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

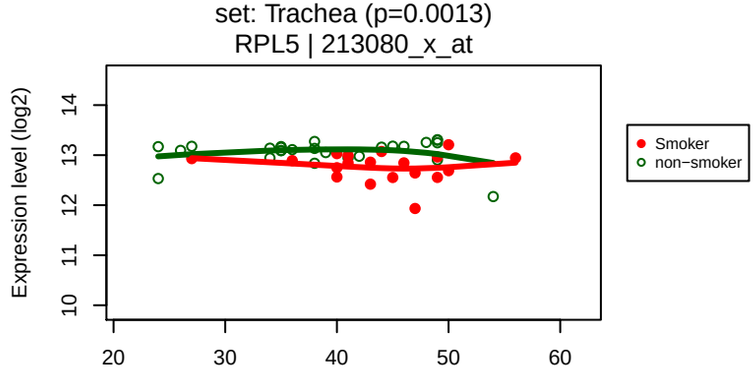

Gene expression ratio in Males
gene: RPL5 | 213080_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8458
p=0.0476
p=0.0018
p=0.3202
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL5 | 213080_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0336
p=0.5191
p=0.0773
p=0.666
SmallAirway2
LargeAirway2
Trachea
Macrophage2

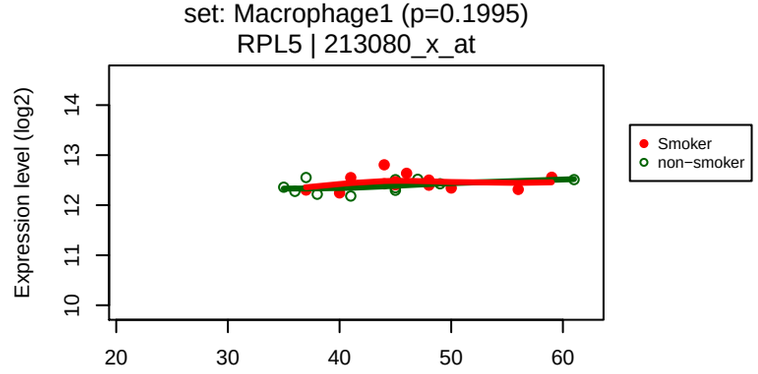

set: SmallAirway1 (p=0.0527)
RPL5 | 213080_x_at
Expression level (log2)
Age
Smoker
non−smoker

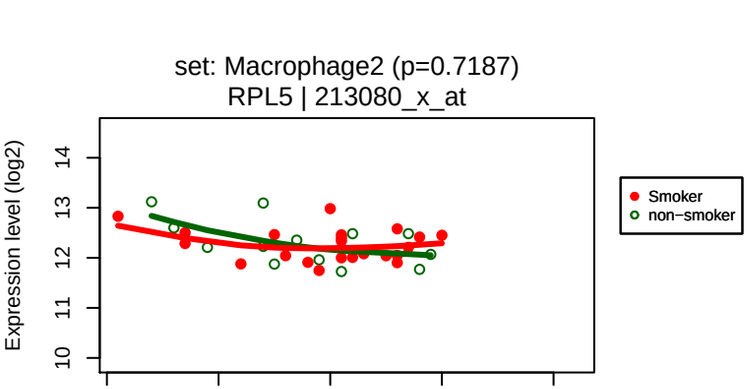

set: SmallAirway2 (p=0.5366)
RPL5 | 213080_x_at
Expression level (log2)
Age
Smoker
non−smoker

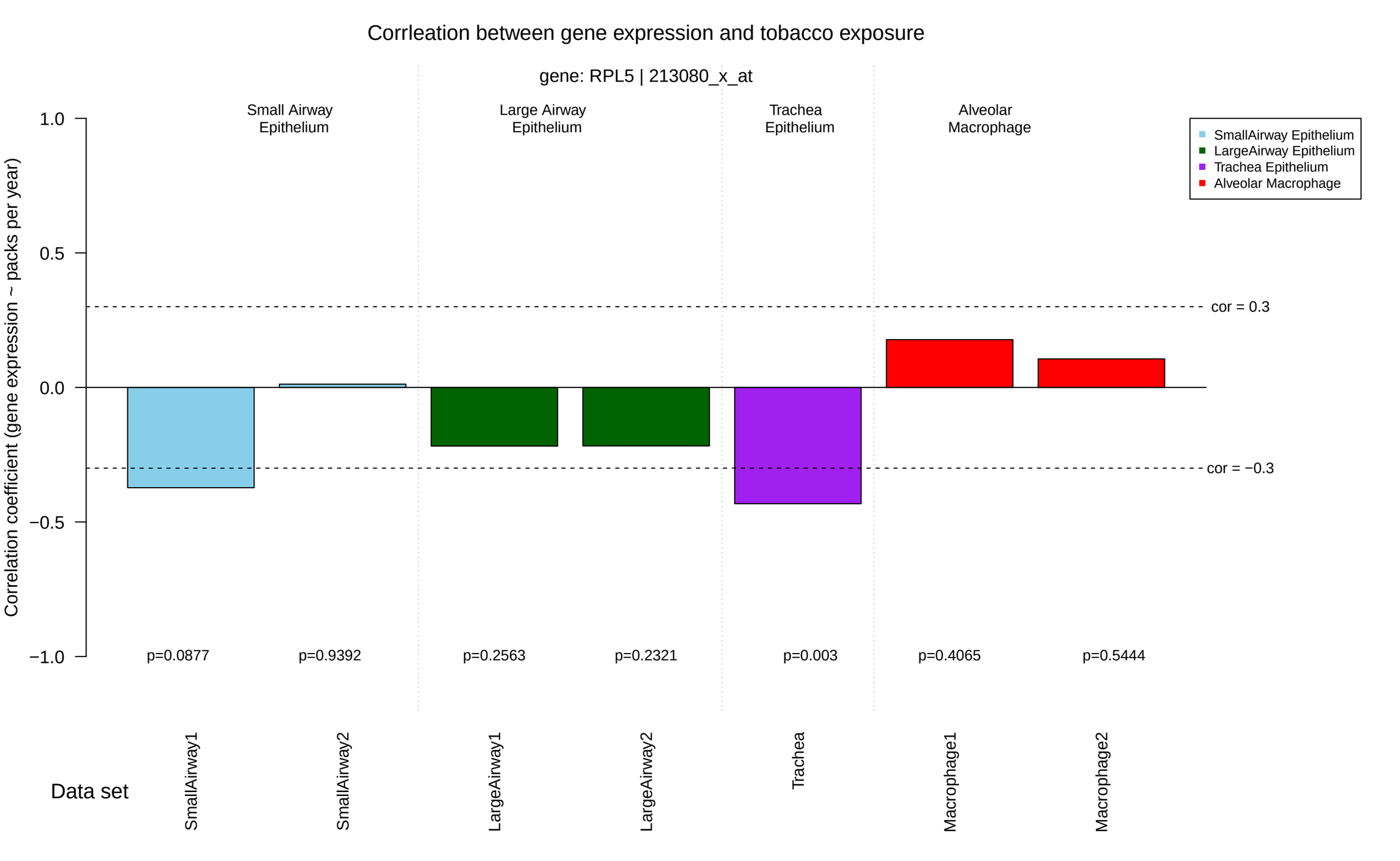

set: LargeAirway1 (p=0.0021)
RPL5 | 213080_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0183)
RPL5 | 213080_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
RPL5 | 213080_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1995)
RPL5 | 213080_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7187)
RPL5 | 213080_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL5 | 213080_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0877
p=0.9392
p=0.2563
p=0.2321
p=0.003
p=0.4065
p=0.5444
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2